



S3.1 The periodic table: classification of elements

Structure 3 · Classification of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The periodic table arranges elements in order of increasing atomic number into periods (rows, corresponding to the outermost occupied principal energy level) and groups (columns, elements with the same number of valence electrons and therefore similar chemical properties), and is divided into s-, p-, d- and f-blocks based on which sub-shell is being filled. Across a period, atomic radius generally decreases and first ionization energy and electronegativity generally increase (due to increasing nuclear charge pulling the same-shell electrons closer, with the same shielding), while down a group atomic radius increases and ionization energy/electronegativity generally decrease (due to additional shielding shells outweighing the increased nuclear charge). Elements are also classified as metals, non-metals or metalloids based on their physical and chemical properties, which correlate with position in the table.

Worked example: follow the reasoning

Explain why the first ionization energy of aluminium is lower than that of magnesium, even though Al has a greater nuclear charge.

Step 1: Mg: $[\text{Ne}]3s^2$; Al: $[\text{Ne}]3s^23p^1$.

Step 2: The electron removed from Mg comes from a filled, lower-energy 3s sub-shell.

Step 3: The electron removed from Al comes from the single 3p electron, which occupies a higher-energy sub-shell than 3s and is slightly further from the nucleus/more shielded by the $3s^2$ electrons.

Answer: Although Al has one more proton than Mg, its outermost electron is in the higher-energy 3p sub-shell (less strongly held, with extra shielding from the $3s^2$ pair), so less energy is needed to remove it, giving Al a lower first ionization energy than Mg despite the general left-to-right increasing trend.

Quick check · multiple choice

Which element is expected to have the largest atomic radius?

- A Na
- B Mg
- C K
- D Ca

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) State and explain the general trend in electronegativity across period 3 (Na to Cl). [3]
(b) The successive ionization energies (in kJ mol^{-1}) of an element X are: 1st = 738, 2nd = 1451, 3rd = 7733, 4th = 10540. Deduce, with reasoning, the group of the periodic table to which X belongs. [2]
(c) Explain why the noble gases (group 18) are generally not assigned an electronegativity value. [1]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) Electronegativity generally increases from Na to Cl across period 3. This is because nuclear charge increases across the period while the number of shielding inner shells stays the same, so the outer electrons (and a bonding pair of electrons) are attracted more strongly by the increasing effective nuclear charge, increasing the atom's ability to attract a shared pair of electrons. [3]

(b) There is a very large jump in ionization energy between the 2nd (1451) and 3rd (7733) values, showing that the 3rd electron removed comes from a much more stable, lower (inner) energy level/full shell. This means the atom has only 2 valence electrons, so X is in group 2. [2]

(c) Noble gases do not readily form bonds/do not share electrons in compounds (they are already stable/have a full outer shell), so there is no meaningful measure of their tendency to attract a shared pair of electrons in a bond. [1]

Quick check answer

C. K is in the same group as Na but one period lower (an extra electron shell), and going down a group increases atomic radius more than the effect of increasing nuclear charge across the period, giving K the largest radius of these four.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

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